Arithmetic D-modules

Daniel Caro

Lecture 1

§ Introduction

Let $X \xrightarrow{f} S$ a smooth morph. of schemes. $\Delta_f : X \hookrightarrow X \times_S X$ the diag morph.

Ex. $S = \operatorname{Spec} \mathbb{C}$, or $\operatorname{Spec} k$, k finite of char $p > 0$.

Def (E.G.A. III, Grothendieck) Choose

$$\begin{tikzcd} & U \underset{\text{open}}{\subset} X \times_S X \arrow[r, shift left, "p_1"] \arrow[r, shift right, "p_2"'] & X \\ X \arrow[ur, hook, "I"] \arrow[ur] & & \end{tikzcd}$$

$P^n_{X/S} := \mathcal{O}_U / I^{n+1}$ sheaf of principal parts of order n

$$D_{X/S,m} = \operatorname{Hom}_{\mathcal{O}_X}(p_{2*} P^m_{X/S}, \mathcal{O}_X)$$

$$D_{X/S} = \bigcup_{m \in \mathbb{N}} D_{X/S,m}$$ sheaf of diff. operators w.r.t. X/S

Local description. Suppose X/S has local coord. $t_1, \dots, t_d$,

$$\begin{tikzcd} X \arrow[rr, "t_1, \dots, t_d", "\text{étale}"'] \arrow[dr] & & \mathbb{A}^d_S \arrow[dl] \\ & S & \end{tikzcd}$$

$$\tau_i = 1 \otimes t_i - t_i \otimes 1$$

$$\left\{ \underline{\tau}^{\underline{k}},\ \underline{k} \in \mathbb{N}^d,\ |\underline{k}| \le m \right\}$$ is a basis as $\mathcal{O}_X$-module of $P^m_{X/S}$

$$\underline{\tau}^{\underline{k}} = \prod_{i=1}^{d} \tau_i^{k_i} \quad \text{if } \underline{k} = (k_1, \dots, k_d)$$

We write $\left\{ \partial^{[\underline{k}]},\ |\underline{k}| \le m,\ \underline{k} \in \mathbb{N}^d \right\}$ the dual basis in $D_{X/S,m}$.

$\{ \partial^{[\underline{k}]} : \underline{k} \in \mathbb{N}^d \}$ a basis of $D_{X/S}$.

When $S = \operatorname{Spec} \mathbb{C}$, $\underline{\partial}^{[\underline{k}]} = \frac{\partial_1^{k_1}}{k_1!} \cdots \frac{\partial_d^{k_d}}{k_d!}$

$= \frac{\underline{\partial}^{\underline{k}}}{\underline{k}!}$

$[k]$: "divided powers", $\partial_i := \underline{\partial}^{[0,\cdots 1,0,\cdots,0]}$

When $S = \operatorname{Spec} \mathbb{F}_p$, $\mathcal{D}_{X/S}$ is not finitely generated as $\mathcal{O}_X$-algebra.

$\mathcal{D}^{(0)}_{X/S}$: $\partial_1, \cdots, \partial_d$ generate this $\mathcal{O}_X$-alg

§1. Divided powers

<u>Def</u>. Let A be an algebra. Let $I \subset A$ be an ideal. By "divided powers" will mean a collection of maps $\gamma_i : I \to A$, $i \in \mathbb{N}$ s.t.

(1) $\forall x \in I$, $\gamma_0(x) = 1$, $\gamma_1(x) = x$, $\forall i > 1$, $\gamma_i(x) \in I$

(2) $\forall x, y \in I$, $\forall i \in \mathbb{N}$, $\gamma_i(x+y) = \sum_{i'+i''=i} \gamma_{i'}(x)\gamma_{i''}(y)$

(3) $\forall \lambda \in A$, $\forall x \in I$, $\forall i \in \mathbb{N}$, $\gamma_i(\lambda x) = \lambda^i \gamma_i(x)$.

(4) $\forall x \in I$, $\forall i, j \in \mathbb{N}$, $\gamma_i(x)\gamma_j(x) = \binom{i+j}{i} \gamma_{i+j}(x)$

(5) $\forall x \in I$, $\forall i \in \mathbb{N}$, $\forall j \geq 1$, $\gamma_i(\gamma_j(x)) = \frac{(ij)!}{i!(j!)^2} \gamma_{ij}(x)$

We say that "γ is a PD str." on I.

"(I, γ) is a PD ideal"

"(A, I, γ) is a PD ring"

<u>Rmk</u>. $k!\,\gamma_k(x) = x^k$.

If $\exists\, m \in \mathbb{N}$ s.t. $mA = 0$, then I is a nil ideal.

<u>Ex</u>. (1) if $I = (0)$, $\gamma_i = 0$.

$\forall I \subset A$ ideal, $\exists!$ PD str.

(2) if A is a $\mathbb{Q}$-alg. $\gamma_k(x) = \frac{x^k}{k!}$

(3) $(p) \subset \mathbb{Z}_{(p)}$, we get a PD str. on (p). $\gamma_i(pa) := a^i \frac{p^i}{i!} \in \mathbb{Z}_{(p)}$

(4) $A = R\langle T_1, \cdots, T_d \rangle$ A is free R-module ($= R[T_1, \cdots, T_d]$ if R is a $\mathbb{Q}$-algebra)

$\uparrow$ $\{ \underline{T}^{[\underline{k}]} : \underline{k} \in \mathbb{N}^d \}$

PD poly. ring

$$\underline{T}^{[\underline{k}]} \underline{T}^{[\underline{k}']} = \binom{\underline{k} + \underline{k}'}{\underline{k}} \underline{T}^{[\underline{k} + \underline{k}']}$$

$$\binom{\underline{k} + \underline{k}'}{\underline{k}} = \prod_{i=1}^{d} \binom{k_i + k_i'}{k_i}$$

$$\underline{T}^{[\underline{k}]} = T_1^{[k_1]} \cdots T_d^{[k_d]}$$

$$\updownarrow$$

$$\frac{T_1^{k_1}}{k_1!}$$

<u>Def</u> Let (A, I, γ) be a PD ring. Let B be an A-alg.

We say that "γ extends to B" when $\exists!$ PD-structure $\bar{\gamma}$ on IB s.t.

$(A, I, \gamma) \to (B, IB, \bar{\gamma})$ is a PD-morph. (This means that we have $A \xrightarrow{\varphi} B$ a morph. of ring, s.t. $\varphi(I) \subset J$, and s.t. $\forall x \in I$, $\varphi(\gamma_i(x)) = \bar{\gamma}_i(\varphi(x))$)

$(A, I) \to (B, J)$ means a $A \to B$ morph. of rings s.t. $\varphi(I) \subset J$

$$\Leftrightarrow \quad \begin{array}{ccc} \operatorname{Spec} B & \longrightarrow & \operatorname{Spec} A \\ \uparrow & \circlearrowright & \uparrow \\ \operatorname{Spec} B/J & \longrightarrow & \operatorname{Spec} A/I \end{array}$$

<u>Ex</u> (1) When B/A is flat, we can always extend PD structure.

(2) When I is principal, it is always possible to extend.

<u>Def</u> Let (A, I, γ) be a PD ring, B an A-algebra, (J, δ) a PD ideal in B.

We say that γ and δ are "compatible" when γ extends to a PD str. $\bar{\gamma}$ on B and $\exists$ PD str. $\bar{\delta}$ on $IB + J$ s.t. $(A, I, \gamma) \longrightarrow (B, IB + J, \bar{\delta})$

$(B, J, \delta) \nearrow$

are PD-morphisms.

§2 PD-envelope

<u>Th</u>. Let (A, I, γ) be a PD ring, B be an A-alg, $I \subset B$ an ideal.

$\exists$ PD ring $(P_\gamma(J), \bar{J}, [\ ])$ a PD ring, s.t.

$$(*) \left[\begin{array}{l} (B, J) \longrightarrow (P_\gamma(J), \bar{J}) \\ [\ ] \text{ and } \gamma \text{ are compatible} \end{array} \right.$$

and $\forall$ (C, K, δ) PD ring satisfying $(*)$, $\exists!$

$$\begin{tikzcd} (P_\gamma(J), \bar{J}, [\ ]) \arrow[rr, dashed, "\exists! \text{ PD mor}"] & \sim & (C, K, \delta) \\ & (B, J) \arrow[ul] \arrow[ur] & \end{tikzcd}$$

We say that $(P_\gamma(J), \bar{J}, [\ ])$ is the PD envelope of J.

<u>Ex</u>. $B = A[T_1, \cdots, T_d]$, $J = (T_1, \cdots, T_d)$,

$P_\gamma(J) = A\langle T_1, \cdots, T_d \rangle$, $\bar{J} = (T_1, \cdots, T_d)$.

$\gamma_i(T_j) = T_j^{[i]}$

<u>Notation</u>. $I^{[m]} :=$ the ideal gen. by $\gamma_{i_1}(x_1) \cdots \gamma_{i_r}(x_r)$, $x_j \in I$

$\sum_{j=1}^{r} i_j \geq m$.

$(I^{[m]})_{m \in \mathbb{N}}$ is the PD filtration of (I, γ).

$P_\gamma^m(J) := P_\gamma(J) / \bar{J}^{[m+1]}$.

$P_\gamma(J) \otimes_{B \text{ flat}} B' \simeq P_\gamma(JB')$

Let S be a scheme. $(\mathcal{O}, \gamma)$ a q.c. PD ideal.

Let $Z \underset{\mathcal{I}}{\hookrightarrow} X$ be a closed immersion of S-schemes

$$\underset{\substack{\text{affine open}\\ \text{subset of } X}}{U} \longmapsto P_r(\Gamma(U, I)) \text{ is a sheaf}$$

$\rightsquigarrow$ $\mathcal{P}_r(I)$ is a q.c. $\mathcal{O}_X$-alg

$$\mathbb{P}_{Z,r}(X) := \operatorname{Spec}(\mathcal{P}_r(I)) \quad , \quad \mathbb{P}^n_{Z,r}(X) := \operatorname{Spec}(\mathcal{P}^n_r(I))$$

Suppose X/S is smooth, $X \underset{\Delta}{\hookrightarrow} X \times_S X$

$$\Delta^n_{X/S,(0)} = \mathbb{P}^n_{X,r}(X \times_S X) = \operatorname{Spec}(\underbrace{\mathcal{P}^n_{X/S,(0)}}_{\text{sheaf of principal parts of level } 0})$$

X/S has local coord $t_1, \dots, t_d$.

a basis of $\mathcal{P}^n_{X/S,(0)}$ is $\{\underline{\tau}^{[\underline{k}]} : |\underline{k}| \le n\}$

$$D^{(0)}_{X/S,n} = \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{P}^n_{X/S,(0)}, \mathcal{O}_X) \quad , \quad \{\underline{\partial}^{\underline{k}} : |\underline{k}| \le n\} \text{ is the dual basis as } \mathcal{O}_X\text{-module.}$$

$D^{(0)}_{X/S}$ is gen'd by $\partial_1, \dots, \partial_d$

$\shortparallel$

the sheaf of diff. ops "of level 0"

$D_{X/S}$ - - - - "of level ∞"

$\mathcal{P}^n_{X/S}$ $\quad$ $\mathcal{P}^n_{X/S,(m)}$ $\quad$ $\mathcal{P}^n_{X/S,(0)}$

$\underline{\tau}^{\underline{k}}$

$$\underline{\tau}^{[\underline{k}]} \longleftrightarrow \frac{\underline{\tau}^{\underline{k}}}{\underline{k}!}$$

Lecture 2. Let $m \in \mathbb{N}$. Partial divided power of level m.

A ring A will be a $\mathbb{Z}_{(p)}$-alg. A PD structure will be compatible with that of $(p) \subset \mathbb{Z}_{(p)}$ $\left(\gamma_i(pa) := \frac{p^i a^i}{i!} \right)$

Def. Let A be a $\mathbb{Z}_{(p)}$-alg, $I \subset A$ an ideal. A "partial divided power of level m structure" on I, is (J, γ), $J \subset I$ ideal, γ is a PD str. on J, s.t. $I^{(p^m)} + pI \subset J$.

$$I^{(p^m)} = \langle x^{p^m} : x \in I \rangle$$

We say that "(I, J, γ) is an m-PD ideal"

"(J, γ) is an m-PD str. on I"

"(A, I, J, γ) is an m-PD ring"

A morph. of m-PD rings $(A, I, J, \gamma) \to (A', I', J', \gamma')$ is a morph. of rings $A \xrightarrow{\varphi} A'$, s.t. $\varphi(I) \subset I'$, $\varphi(J) \subset J'$, $\forall x \in J$, $\forall i \in \mathbb{N}$, $\varphi(\gamma_i(x)) = \gamma_i'(\varphi(x))$.

Notation Let (A, I, J, γ) be an m-PD ring. Let $k \in \mathbb{N}$. $k = p^m q + r$, $0 \le r < p^m$.

$k' \le k$, $k'' = k - k'$.

$$\left\{ \begin{matrix} k \\ k' \end{matrix} \right\}_{(m)} := \frac{q!}{q'!\, q''!}$$

$$\left\langle \begin{matrix} k \\ k' \end{matrix} \right\rangle_{(m)} = \binom{k}{k'} \Big/ \left\{ \begin{matrix} k \\ k' \end{matrix} \right\}_{(m)} \in \mathbb{Z}_{(p)}.$$

e.g. $\left\{ \begin{matrix} k \\ k' \end{matrix} \right\}_{(0)} = \binom{k}{k'}$, $\left\langle \begin{matrix} k \\ k' \end{matrix} \right\rangle_{(0)} = 1$.

$\forall x \in I$, $\forall k \in \mathbb{N}$, $k = p^m q + r$, $0 \le r < q^m$. $x^{\{k\}_{(m)}} = x^r \gamma_q(x^{p^m})$ plays the role of $\frac{x^k}{q!}$.

When $m=0$, $x^{\{k\}(m)} \sim \frac{x^k}{k!}$

"$m=\infty$", $x^{\{k\}(\infty)} \sim x^k$.

$$q!\, x^{\{k\}(m)} = x^k$$

From the properties satisfied by divided powers, we get:

(i) $\forall x \in I$, $x^{\{0\}} = 1$, $x^{\{1\}} = x$, $\forall i \geq 1$, $x^{\{i\}} \in I$

$\forall i \geq p^m$, $x^{\{i\}} \in J$

(ii) $\forall x \in I$, $\forall a \in A$, $\forall k \in \mathbb{N}$, $(ax)^{\{k\}} = a^k x^{\{k\}}$.

(iii) $\forall x, y \in I$, $\forall k \in \mathbb{N}$, $(x+y)^{\{k\}} = \sum_{k'+k''=k} \left\langle {k \atop k'} \right\rangle_{(m)} x^{\{k'\}} y^{\{k''\}}$

(iv) $\forall x \in I$, $\forall k, k' \in \mathbb{N}$, $x^{\{k\}} x^{\{k'\}} = \left\{ {k+k' \atop k'} \right\}_{(m)} x^{\{k+k'\}}$

(v) $\forall x \in I$, $\forall k', k'' \in \mathbb{N}$. $\left(x^{\{k'\}}\right)^{\{k''\}} = \frac{q!}{q''!\,(q'!)^{q}} x^{\{k\}}$

$k = k' k''$

$= p^m q + r$

$q' = [k'/p^m]$

$q'' = [k''/p^m]$

<u>Example</u> A is R-algebra

$A = R\langle T_1, \cdots, T_d \rangle_{(m)}$ m-PD polynomial ring

a basis is $\underline{T}^{\{\underline{k}\}(m)}$, $\underline{k} \in \mathbb{N}^d$

$I = (T_1, \cdots, T_d)$ has a canonical m-PD structure.

$$\underline{T}^{\{\underline{k}\}} \cdot \underline{T}^{\{\underline{k}'\}} = \left\{ {\underline{k}+\underline{k}' \atop \underline{k}} \right\}_{(m)} \underline{T}^{\{\underline{k}+\underline{k}'\}}$$

$$\| $$

$$\prod_{i=1}^{d} \left\{ {k_i + k_i' \atop k_i} \right\}_{(m)}$$

$$J = \bigoplus_{\underline{k} \in \mathbb{N}^d} R\, \underline{T}^{\{\underline{k}\}(m)}$$

$\exists i$, $p^m \mid k_i$

By replacing everywhere "PD" by "m-PD" we define the m-PD envelope.

If B is an A-alg, $J \subset B$ ideal, $P_\gamma(J)_{(m)}$ is the m-PD envelope of J compatible with γ.

<u>Example</u>: $B = A[T_1, \cdots, T_d]$, the m-PD envelope is $A\langle T_1, \cdots, T_d\rangle_{(m)}$.

Let S be a scheme over $\mathbb{Z}_{(p)}$, $(\mathcal{O}, \square, \gamma)$ q.c. m-PD sheaf

Let $Z \overset{u}{\hookrightarrow} X$ a closed immersion of S-schemes.

$$\underset{Z,(m),\gamma}{P}(X) = P_{(m),\gamma}(u)$$

is the m-PD envelope of u.

$$\{\text{closed } S\text{-immersion}\} \longrightarrow \{\text{m-PD closed immersion compatible with } \gamma\}$$

$$u \longmapsto P_{(m),\gamma}(u)$$

is a left adjoint of the inclusion of cat.

if $m' \geq m$,

$$\begin{array}{ccc} I^{(p^m)} + pI & \subset & J \\ \cup & & \subset \\ I^{(p^{m'})} + pI & & \end{array}$$

$$P_{(m),\gamma}(u) \longrightarrow P_{(m'),\gamma}(u)$$

We have a m-PD filtration

$$(I^{\{n\}})_{n \in \mathbb{N}}$$

$$P^n_{(m),\gamma}(J) = P_{(m),\gamma}(J) / \overline{J}^{\{n+1\}}$$

Let X/S be smooth of $\mathbb{Z}_{(p)}$-schemes.

$$\Delta_{X/S} : X \to X \times_S X$$

$\Delta^m_{X/S}$: the m-PD envelope of $\Delta_{X/S}$

$\overset{\shortparallel}{\mathrm{Spec}}\left(\mathcal{P}^n_{X/S,(m)}\right)$, sheaf of principal parts of level m.

Locally, X/S has local coord. $t_1, \dots, t_d$

$$\begin{tikzcd} X \arrow[rr, "t_1, \dots, t_d", "\text{ét}"'] \arrow[dr] & & \mathbb{A}^d_S \arrow[dl] \\ & S & \end{tikzcd}$$

$\tau_i = 1 \otimes t_i - t_i \otimes 1$

A basis of $P^n_{X/S,(m)}$ as $\mathcal{O}_X$-module is $\underline{\tau}^{\{\underline{k}\}_{(m)}}$, $|\underline{k}| \leq n$

$\underline{\tau}^{\{\underline{k}\}_{(m)}} = \prod_{i=1}^{d} \tau_i^{\{k_i\}_{(m)}}$, $|\underline{k}| = k_1 + \dots + k_d$

$$\mathcal{D}^{(m)}_{X/S,n} := \mathcal{H}om_{\mathcal{O}_X}(\underset{\text{left}}{\uparrow} P^n_{X/S,(m)}, \mathcal{O}_X)$$

$$\mathcal{D}^{(m)}_{X/S} := \bigcup_{n \in \mathbb{N}} \mathcal{D}^{(m)}_{X/S,n}$$: sheaf of diff. ops of level n.

When X/S has local coord., we write $\underline{\partial}^{\langle \underline{k} \rangle_{(m)}} \longleftrightarrow q! \, \underline{\partial}^{[\underline{k}]} \longleftrightarrow q! \frac{\underline{\partial}^{\underline{k}}}{\underline{k}!}$ the dual basis of $\underline{\tau}^{\{\underline{k}\}_{(m)}}$

$$\begin{tikzcd} P^n_{X/S} \otimes_{\mathcal{O}_X} P^{n'}_{X/S} & \mathcal{O}_{X \times_S X}/I^{n+n'+1} = P^{n+n'}_{X/S} \arrow[l, "\delta^{n,n'}"] \\ (a \otimes 1) \otimes (1 \otimes b) \arrow[d, mapsto] & a \otimes b \arrow[l, mapsto] \arrow[d, mapsto] \\ P^n_{X/S,(m)} \otimes P^{n'}_{X/S,(m)} & P^{n+n'}_{X/S,(m)} \arrow[l, "\delta^{n,n'}_{(m)}"] \end{tikzcd}$$

$$P^n_{X/S,(\infty)} \to \cdots \to P^n_{X/S,(m)} \to \cdots \to P^n_{X/S,(0)}$$
$$P^n_{X/S,(\infty)} = P^n_{X/S}$$

$P \in \mathcal{D}^{(m)}_{X/S,n}$

$P' \in \mathcal{D}^{(m)}_{X/S,n'}$, $PP' \in \mathcal{D}^{(m)}_{X/S,n+n'}$

$$PP': P^{n+n'}_{X/S,(m)} \xrightarrow[\delta^{n,n'}_{(m)}]{} P^n_{(m)} \otimes_{\mathcal{O}_X} P^{n'}_{(m)} \xrightarrow[\mathrm{id} \otimes P']{} P^n_{(m)} \xrightarrow[P]{} \mathcal{O}_X$$

We get a ring structure.

We have an action $\mathcal{O}_X$

if $P \in \mathcal{D}^{(m)}_{X/S,n}$, $a \in \mathcal{O}_X$.

$$\begin{array}{ccccc} \mathcal{O}_X & \longrightarrow & \mathcal{P}^n_{X/S,(m)} & \xrightarrow{p} & \mathcal{O}_X \\ a & \xmapsto{d_1} & \text{"}1 \otimes a\text{"} & \longmapsto & P(a) \end{array}$$

We have some formulae

- $\forall\, \underline{k}, \underline{k}'' \in \mathbb{N}^d$, $\underline{\partial}^{\langle \underline{k} \rangle} \cdot \underline{\partial}^{\langle \underline{k}' \rangle} = \binom{k+k'}{k} \underline{\partial}^{\langle k+k' \rangle}$

- $\forall\, f \in \mathcal{O}_X$, $\forall\, k \in \mathbb{N}^d$,

$$\underline{\partial}^{\langle \underline{k} \rangle} \cdot f = \sum_{\underline{k}' + \underline{k}'' = \underline{k}} \left\{ \begin{matrix} \underline{k} \\ \underline{k}' \end{matrix} \right\}_{(m)} \underline{\partial}^{\langle \underline{k}' \rangle}(f)\, \underline{\partial}^{\langle \underline{k}'' \rangle}$$

$\mathrm{gr}\, \mathcal{D}^{(m)}_{X/S} = \bigoplus \mathcal{D}^{(m)}_{X/S,n} \Big/ \mathcal{D}^{(m)}_{X/S,n+1}$ is commutative

---

<u>Lecture 3</u>. X/S smooth morph of $\mathbb{Z}_{(p)}$-schemes

with local coord. $t_1, \cdots, t_d$, then

$\mathcal{D}^{(m)}_{X/S}$ is gen. as $\mathcal{O}_X$-alg by $\{ \partial_i^{\langle p^n \rangle} : 1 \le n \le m,\ \forall\, i = 1, \cdots, d \}$

<u>Cohomological operations</u>

- Extraordinary inverse image

$$\begin{tikzcd} \Delta^n_{Y/T,(m)} \arrow[r, shift left, "q_0"] \arrow[r, shift right, "q_1"'] \arrow[d, dashed] & Y \arrow[r, "\text{smooth}"] \arrow[d, "f"] \arrow[dl, phantom, "\circlearrowright"] & T \arrow[d, "L"] \\ \Delta^n_{X/S,(m)} \arrow[r, shift left] \arrow[r, shift right] \arrow[rr, bend right, "p_1^n"'] & X \times_S X \arrow[r] & S \end{tikzcd}$$

p_0^n smooth

$$\begin{tikzcd} X \arrow[r, hook] \arrow[dr] & X \times_S X \times_S X \\ & \Delta^n_{X/S,(m)}(2) \arrow[u] \arrow[r, shift left, "p_{01}"] \arrow[r, shift right, "p_{02}"'] \arrow[r, shift right, "p_{12}"'] & \Delta^n_{X/S,(m)} \end{tikzcd}$$

FACT Let $\mathcal{E}$ be an $\mathcal{O}_X$-module. To get a structure of left $D^{(m)}_{X/S}$-mod extending the $\mathcal{O}_X$-mod str. is equiv. to

get some compatible isomorphism

$$\mathcal{E}_n : p_1^{n*}(\mathcal{E}) \xrightarrow{\sim} p_0^{n*}(\mathcal{E})$$

s.t. $\mathcal{E}_0 = id$ satisfying the cocycle condition

$$p_{02}^{n*}(\mathcal{E}_n) = p_{01}^{n*}(\mathcal{E}_n) \circ p_{12}^{n*}(\mathcal{E}_n)$$

m-PD stratification

If $\mathcal{E}$ is left $D^{(m)}_{X/S}$-mod $\rightsquigarrow$ $(\mathcal{E}, \mathcal{E}_n)$

$f^*(\mathcal{E})$ get $\left((f\times f)^*(\mathcal{E}, \mathcal{E}_n)\right)$

$\rightsquigarrow$ $D^{(m)}_{Y/T}$-mod str. on $f^*(\mathcal{E})$.

Suppose $T = S$.

$D^{(m)}_{Y\to X} := f^*\left(D^{(m)}_{X/S}\right)$: $\left(D^{(m)}_{Y/S}, f^{-1}D^{(m)}_{X/S}\right)$-bimodule

$$f^! : D^b_{qc}\left(D^{(m)}_{X/S}\right) \longrightarrow D^b_{qc}\left(D^{(m)}_{Y/S}\right)$$

$$\mathcal{E} \longmapsto D^{(m)}_{Y\to X} \overset{\mathbb{L}}{\underset{f^{-1}D^{(m)}_{X/S}}{\otimes}} f^{-1}\mathcal{E}\,[d_Y - d_X] =: f^!(\mathcal{E})$$

$\omega_{X/S} := \Lambda^d \Omega^1_{X/S}$ has a canonical right $D^{(m)}_{X/S}$-structure.

- Suppose X/S has local coord $t_1, \dots, t_d$.

$$\omega_{X/S} = \mathcal{O}_X\, dt_1 \wedge \dots \wedge dt_d$$

Let $P = \sum_{\underline{k}\in\mathbb{N}^d} \underset{\in \mathcal{O}_X}{a_{\underline{k}}}\, \underline{\partial}^{\langle \underline{k}\rangle} \in D^{(m)}_{X/S}$, $\forall a \in \mathcal{O}_X$,

$\underline{\partial}^{\langle \underline{k}\rangle} \Leftrightarrow \frac{q!}{k!}\underline{\partial}^{\underline{k}}$

$$(a\, dt_1 \wedge \cdots \wedge dt_d) \cdot P = {}^tP(a)\, dt_1 \wedge \cdots \wedge dt_d$$

$$ {}^tP = \sum_{\underline{k} \in \mathbb{N}^d} (-1)^{|\underline{k}|}\, \underline{\partial}^{\langle \underline{k} \rangle}\, a_{\underline{k}}$$

$\forall\ P, Q \in D^{(m)}_{X/S}$, ${}^t(PQ) = {}^tQ\, {}^tP$

$$- \underset{\mathcal{O}_X}{\otimes} \omega_X : \left\{ \text{left } D^{(m)}_{X/S}\text{-mod} \right\} \xrightarrow{\sim} \left\{ \text{right } D^{(m)}_{X/S}\text{-mod} \right\}$$

$$\xleftarrow[\sim]{} \quad - \underset{\mathcal{O}_X}{\otimes} \omega_X^{-1}$$

$f : Y \xrightarrow{\text{qcqs}} X$, $Y \xrightarrow{\text{smooth}} S \xleftarrow{\text{smooth}} X$, S noeth of finite Krull dim.

$$D^{(m)}_{X \leftarrow Y} := f^*\left(D^{(m)}_{X/S} \underset{\mathcal{O}_Y}{\otimes} \omega_X^{-1}\right) \underset{\mathcal{O}_Y}{\otimes} \omega_Y \quad : \left(f^{-1} D^{(m)}_{X/S},\ D^{(m)}_Y\right)\text{- bimodule}$$

$$f_+ : D^b_{qc}(D^{(m)}_Y) \longrightarrow D_{qc}(D^{(m)}_X)$$

$$\mathcal{F} \longmapsto \mathbb{R} f_*\left(D^{(m)}_{X \leftarrow Y} \overset{\mathbb{L}}{\underset{D^{(m)}_Y}{\otimes}} \mathcal{F}\right)$$

Ex. if f is smooth, $f_+(\mathcal{E}) = \mathbb{R} f_*\left(\mathcal{E} \underset{\mathcal{O}_Y}{\otimes} \Omega^{\cdot}_{Y/X}\right)[d_{Y/X}]$

$D^{(m)}_{X/S}$ is coherent, of finite cohomological dim.

$$D^b_{perf}(D^{(m)}_{X/S}) = D^b_{coh}(D^{(m)}_{X/S})$$

$$\mathbb{D} : D^b_{coh}(D^{(m)}_{X/S}) \circlearrowleft$$

$$\mathcal{E} \longmapsto \mathbb{R}\mathcal{H}om_{D^{(m)}_{X/S}}\left(\mathcal{E}, D^{(m)}_{X/S}\right) \underset{\mathcal{O}_X}{\otimes} \omega_X^{-1}[d_X]$$

p-adic cohomology over quasi-proj. var. of char $p>0$.

Let V be a DVR of mixed char $(0,p)$, $p>0$.

π unif., $k = V/\pi$ perfect, $K = \mathrm{Frac}(V)$, $S = \mathrm{Spf}\, V$, $S_i = \mathrm{Spec}(V/\pi^{i+1})$

$$\mathcal{P} = \widehat{\mathbb{P}}^n_V, \quad P_i = \mathbb{P}^n_{S_i}, \quad \mathcal{P} = \lim_{i\to+\infty} P_i \qquad S = S_0,\ P = P_0$$

We put $\hat{D}^{(m)}_{\mathcal{P}} := \varprojlim_i D^{(m)}_{P_i/S_i}$

When $\mathcal{P}/S$ has local coord $t_1, \dots, t_d$.

$$\Gamma(\mathcal{P}, \hat{D}^{(m)}_{\mathcal{P}}) = \left\{ \sum_{\underline{k}\in\mathbb{N}^d} \underbrace{a_{\underline{k}}}_{\in \mathcal{O}_{\mathcal{P}}} \partial^{\langle \underline{k}\rangle_{(m)}} : \lim_{|\underline{k}|\to\infty} a_{\underline{k}} \to 0 \right\}$$

$$\begin{tikzcd} \hat{D}^{(m)}_{\mathcal{P}} \arrow[r, hook] \arrow[d] & \hat{D}^{(m+1)}_{\mathcal{P}} \arrow[d] \\ D^{(m)}_{P} \arrow[r] & D^{(m+1)}_{P} \end{tikzcd}$$

$$\hat{D}^{(m)}_{\mathcal{P}} \longrightarrow \hat{D}^{(m+1)}_{\mathcal{P}}$$

$$\partial^{\langle \underline{k}\rangle_{(m)}} \mapsto \frac{q!}{q'!}\partial^{\langle \underline{k}\rangle_{(m+1)}}$$

$$\underline{k} = p^m q + \underline{r}$$

$$\underline{k} = p^{m+1} q' + \underline{r}'$$

$$\hat{D}_{\mathcal{P}} \supset D^{\dagger}_{\mathcal{P}} := \bigcup_{m\in\mathbb{N}} \hat{D}^{(m)}_{\mathcal{P}}$$

"$\dagger$": weak completion (of $D_{\mathcal{P}}$)

$\cap$ $\hat{D}_{\mathcal{P},\mathbb{Q}}$ $\cap$

$\mathcal{P}$ loc. coord. $t_1, \dots, t_d \rightsquigarrow \partial_1, \dots \partial_d$

$p \in \Gamma(\mathcal{P}, D^{\dagger}_{\mathcal{P}}) \iff$ we write it $p = \sum_{\underline{k}\in\mathbb{N}^d} a_{\underline{k}} \partial^{[\underline{k}]}$, $a_{\underline{k}} \in \mathcal{O}_{\mathcal{P}}$

$$\boxed{\exists\, c, \eta,\ \eta<1,\ \|a_{\underline{k}}\| \le c\eta^{|\underline{k}|}}$$

$\hat{D}_{\mathcal{P}}$ $\quad \sum \underset{\substack{\downarrow \\ 0}}{a_{\underline{k}}} \partial^{[\underline{k}]}$ when $|\underline{k}| \to +\infty$

$D^b_{hol}(\mathcal{D}^\dagger_{P,\mathbb{Q}})$ Let $\mathcal{E}$ be a coh. $\mathcal{D}^\dagger_{P,\mathbb{Q}}$-mod.

$\mathcal{E}$ is holonomic iff $\forall i \neq d_P = n$,

$$\mathcal{H}^i \underbrace{\mathbb{D}(\mathcal{E})}_{\text{dual as } \mathcal{D}^\dagger\text{-module}} = 0$$

$F\text{-}D^b_{hol}(\mathcal{D}^\dagger_{P,\mathbb{Q}}) = \{(\mathcal{E},\phi) : \mathcal{E} \in D^b_{hol}(\mathcal{D}^\dagger_{P,\mathbb{Q}}) + \phi : F^*\mathcal{E} \xrightarrow{\sim} \mathcal{E}\}$ in $D^b(\mathcal{D}^\dagger_{P,\mathbb{Q}})$

coefficients over $P = \mathbb{P}^n_k$

$U = \hat{\mathbb{A}}^1_V$, $\quad p \longmapsto p'(t)$

$$\Gamma(U, \mathcal{O}_{U,\mathbb{Q}}) = K\{t\} \xrightarrow{\nabla} K\{t\}$$

$$= \{\sum_{k\in\mathbb{N}} \lambda_k t^k : \lambda_k \to 0 \text{ when } k \to +\infty\}$$

coker $\nabla \neq 0$

$$\int \sum_{n\in\mathbb{N}} p^n t^{p^n-1} = \sum t^{p^n} \notin K\{t\}$$

$$V[t]^\dagger \otimes_V K \xrightarrow{\nabla} V[t]^\dagger \otimes_V K$$

$$\| $$

$$\{\sum_{k\in\mathbb{N}} a_k t^k : \exists\, c, \eta, \eta<1, \|a_k\| \le c\eta^k\}$$

coker $\nabla = 0$

If $T \subset P$ is a divisor

$\mathcal{O}_P({}^\dagger T)$ — left $\mathcal{D}^\dagger_P$-mod

locally $T = V(\bar{a})$, $a \in \mathcal{O}_P$, $\mathcal{O}_P({}^\dagger T)_{\mathbb{Q}} = \mathcal{O}_P[\frac{1}{a}]^\dagger \otimes_V K$

$\mathcal{O}_{\mathfrak{P}}({}^{\dagger}T)_{\mathfrak{A}} \longleftrightarrow \mathcal{O}_{\mathfrak{U}}$

$\mathfrak{U} = \mathfrak{P} \setminus T$. $\quad \mathcal{O}_{\mathfrak{P}}({}^{\dagger}T)_{\mathfrak{A}}$ is coh as $\mathcal{D}^{\dagger}_{\mathfrak{P},\mathfrak{A}}$-mod

$$({}^{\dagger}T) \text{ "=" } \mathcal{O}_{\mathfrak{P}}({}^{\dagger}T) \overset{\text{"}\dagger\text{"}}{\underset{\mathcal{O}_{\mathfrak{P}}}{\otimes}} (-)$$

$$\| $$

$$\varinjlim \hat{\mathcal{B}}^{(m)}(T)$$

- If $Z \hookrightarrow \mathfrak{P}$ is a closed subvariety,

$({}^{\dagger}Z)$

Let Y be a q-proj. variety $/k$.

$$\begin{array}{ccccc} Y & \hookrightarrow & \mathbb{P}^n_k & \not\hookrightarrow & \mathfrak{P} = \hat{\mathbb{P}}^n_V \\ \text{open} \cap & & \| & & \\ \bar{Y} & \not\hookrightarrow & P & & \\ & \text{closed} & & & \end{array}$$

$Y \rightsquigarrow$ the full subcat. of $F\text{-}D^b_{hol}(\mathcal{D}^{\dagger}_{\mathfrak{P},\mathfrak{A}})$ of complexes $\mathcal{E}$ s.t. $\mathcal{E}$ has supp. in $\bar{Y}$

and s.t. $({}^{\dagger}\bar{Y}\setminus Y)(\mathcal{E}) \simeq \mathcal{E}$. $\qquad \mathcal{E} \simeq \mathbb{R}\Gamma^{\dagger}_{Y}\mathcal{E}$

This is the p-adic analogue of $D^b_{hol}(\mathcal{D}_Y)$

$$\begin{tikzcd} Y \arrow[r, hook] \arrow[dr, hook] & \mathfrak{P} & \mathfrak{P}\times\mathfrak{P}' \arrow[l, "\pi_1"'] \arrow[d, "\pi_2"] \\ & & \mathfrak{P}' \end{tikzcd}$$

$$\begin{tikzcd} p' \arrow[r, "f"] & p \\ Y' \arrow[u] \arrow[r, "f"'] \arrow[ur, phantom, "\circlearrowright"] & Y \arrow[u] \end{tikzcd}$$

$$f_+ \underset{\text{def}}{=} f_+ : D(Y') \to D(Y)$$

$$f^! : D(Y) \to D(Y')$$

$$\mathcal{E} \longmapsto \mathbb{R}\Gamma^+_{Y'} f^!(\mathcal{E})$$

$$\mathbb{D}_Y : D(Y) \to D(Y)$$

$$\mathcal{E} \longmapsto \mathbb{R}\Gamma^+_Y \mathbb{D}(\mathcal{E})$$

$$- \overset{+}{\underset{\mathcal{O}_Y}{\otimes}} -$$

$$f_! = \mathbb{D} f_+ \mathbb{D}$$

$$f^+ = \mathbb{D} f^! \mathbb{D}$$

Y smooth. $\quad F\text{-}\mathrm{Isoc}^+(Y/K)$